WATERWATCH · RISK REGISTER AND HAZARD ANALYSIS

WaterWatch risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: project risks, preliminary hazard analysis and first-pass design FMEA for option A

WaterWatch risk register and hazard analysis

The top risks are that the low-cost chlorine sensor drifts faster than a monthly calibration can absorb, so alerts go wrong in both directions, and that people read a quiet sensor as proof the water is safe. The largest hazards are contamination of the supply through the sample line, the lithium cell in a hot enclosure, and the pole. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be found) to 5 (unlikely to be found).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
P1	Chlorine drift beyond the monthly allowance	Membrane-free graphite sensor and pseudo-reference drift (WWT-STM-001 S12, S14)	False alarms or missed loss of residual; operators stop trusting alerts	4	5	Design lead	Bench drift test is the first TRL 4 task; fallback is option B's DPD chemistry or a membrane probe at higher cost
P2	No partner adopts it	Water Mission not yet approached; operators may prefer proven commercial analysers	No pilot sites, no field data	3	4	Project lead	Approach the first candidate with this TRL 3 pack; offer co-design of thresholds and alerts
P3	Cost above the value-engineering target	USD 325 against USD 300; recurring airtime	Fewer points monitored per budget	3	2	Design lead	LoRaWAN variant where a gateway exists (about USD 304); bulk fixings and plates
P4	Network coverage too weak at the site	Rural LTE-M or 2G gaps	Late or lost alerts	3	3	Scheme operator	Survey signal at each site; pole-top antenna or LoRaWAN variant (WWT-DDR-003 A2)
P5	Vandalism or theft	Panel, battery and modem are of value	Loss of the unit	3	3	Scheme operator	Tamper-resistant shield screws on field units (WWT-DEC-001); community ownership
P6	Type approval delays	Radio equipment rules in the chosen country (WWT-REG-001 R7)	No legal field installation	3	3	Electronics lead	Use a modem module already approved in the country
P7	Sun shield less effective than assumed	Shield factor of 0.25 assumed (S5)	Battery too hot to charge on hot days	3	3	Design lead	Log enclosure temperature at TRL 4; larger gap or a second reflective layer
P8	Lightning, static and condensation	Exposed pole and panel; daily temperature swings in a sealed box	Board or modem damage; corrosion	2	3	Electronics lead	Pole bonded to the footing; surge protection on the panel and antenna leads; pressure-equalising vent and a desiccant pack; to assess at TRL 4 (independent review, 2026-10-25)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Contamination of the drinking water supply	Backflow from the cell or drain into the riser; unsafe wetted materials	Illness of people who drink from the tap	5	1	Isolation valve and two check valves in series (supply train and cell inlet, WWT-DEC-001 F8) with the normally closed valve; open air break at the cell outlet so drain water cannot reach the cell (PR-014, WWT-STM-001 S8, S30); food-contact grades upstream of the check valve	Low
H2	Reading taken as proof the water is safe	A normal reading does not cover pathogens or chemicals	Illness if lab testing stops	4	3	Alerts state measurements only (PR-013); documentation and training say laboratory testing stays necessary	Medium
H3	Lithium cell overheats or is overcharged	Hot enclosure, charger fault, damaged cell	Burns, fire in the enclosure	4	2	LiFePO4 chemistry; protection board, fuse at the cell, 0 to 45 °C charge window (PR-016); ventilated sun shield; UN 38.3 cells	Low
H4	Pole falls or is struck	Poor footing, high wind, impact	Injury to people at the tapstand	4	1	300 x 600 mm footing cast on the survey visit, factor 4.3 in dry sand and 2.2 saturated in a 35 m/s gust (S17); pole stress factor 4.0 (S16)	Low
H5	Sharp edges and pinch points	Cut plates, U-bolt ends, shield edges	Cuts to caretakers and children	2	3	Edges broken on every cut part (model, build plan); U-bolt ends inside the shield	Low
H6	Chemical exposure during calibration	DPD tablets; turbidity standards	Skin or eye irritation; toxic precursors if formazin is made locally	3	2	Prepared, stabilised standards only; gloves; supplier safety data sheets (S24)	Low
H7	Flushed water mistaken for a separate water point or pooling	Drain hose discharge	Slips; people collecting flush water	2	2	Drain to the tapstand basin or a soakaway agreed with the committee (WWT-DDR-001 D7)	Low
H8	Working on a pressurised riser	Fitting the saddle tee	Water jet, injury	2	2	Shut off the supply or use a tapping tool; build plan safety stops	Low

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Free chlorine sensor (BOM 11)	Read free chlorine to PR-001	Slope drifts between visits	Wrong alerts (missed or false)	Electrode fouling, pseudo-reference drift, flow and conductivity effects (S32, S36)	5	4	3	60	Drift test at TRL 4; monthly DPD comparison flags a large error; consider a weekly self-check
F2	Turbidity optics (BOM 10)	Read turbidity to PR-002	Window fouled	Reading high, false turbidity alerts	Biofilm, silt on windows	3	3	2	18	Monthly wipe; 180 degree reference detector shows window loss
F3	Latching valve (BOM 13)	Open and close the flush	Sticks open	Water wasted (up to 720 L/ day), cell over-flushed	Debris, scale	3	2	2	12	Strainer upstream; daily water count from flush time; alert on unexpected flow
F4	Latching valve (BOM 13)	Open and close the flush	Fails to open	No fresh sample; stale readings	Coil fault, low supply pressure	4	2	3	24	Detect by unchanged temperature and chlorine over several readings; fault alert
F5	Pressure-compensating regulator (BOM 14)	Hold 0.5 L/min	Flow low below 0.7 bar	Stale fraction rises to 8.5 % (WWT-CAL-001 D5)	Low supply pressure	3	3	3	27	Site pressure survey; longer flush set per site
F6	Check valves and air break (BOM 14, 15)	Protect the supply	One check valve leaks back	The second check valve still holds (F8); air may enter the sample line	Seat wear	5	1	4	20	Two check valves in series; air break kept open by design; inspect at the monthly visit
F7	Battery and charger (BOM 5, 6)	Store energy	Charging blocked on hot days	Autonomy drains in heat waves	Enclosure above 45 °C	3	3	2	18	Sun shield; low-battery alert; log temperature at TRL 4
F8	Cellular modem and antenna (BOM 7, 8)	Upload and send SMS	No network	Alerts late or lost	Coverage, SIM, antenna damage	4	3	2	24	Three tries; 90 days on board; site survey; LoRaWAN variant
F9	Enclosure and glands (BOM 3, 16)	Keep water out (IP66)	Water ingress at a gland	Corrosion, board failure	Gland not tightened, cable too small for the gland	4	2	3	24	Glands matched to cable diameters; check at commissioning;

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
										pressure-equalising vent
F10	Flow cell lid and gasket (BOM 9)	Hold water in the cell	Leak at the lid	Air in the cell, false turbidity, wet electronics below	Thumb nuts loose, gasket aged	3	2	2	12	Thumb nuts checked at each visit; spare gasket in the caretaker kit
F11	Pole and footing (BOM 1)	Hold the unit	Pole leans	Panel and enclosure misaligned; risk H4	Footing undersized for soft or wet soil	4	1	2	8	Factor 2.2 in saturated sand (WWT-CAL-001 I4); clay and made ground checked at the survey visit; larger footing where needed
F12	Flow cell (BOM 9)	Hold water	Water freezes in the cell	Cracked cell or lid, leak	Clear nights near 0 °C (WWT-STM-001 S33)	3	2	3	18	Drain the cell for the cold season at sites that reach 0 °C; insulate; to assess with the partner
F13	Sample line (BOM 14)	Carry the sample	Second check valve sticks shut	No sample; stale readings	Debris past the strainer	3	2	3	18	Same detection as F4; strainer upstream; check at the monthly visit

The highest RPN is the chlorine sensor's drift (F1, 60), the same open question as P1 and WWT-STM-001 S12.